DESIGN OF GEOMINER 1 : A HYBRID VTOL UAV FOR INTELLIGENT MINERAL EXPLORATION AND MINING OPERATIONS



PROJECT PROPOSAL — GROUP FOUR
UAV SYSTEMS ENGINEERING
INSA TALENT CENTER · DEPARTMENT OF AEROSPACE
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SCOPE

A PROFESSIONAL HYBRID VTOL UAV SYSTEM ENGINEERED FOR MINERAL EXPLORATION, GEOLOGICAL MAPPING, MINE-SITE SURVEYING, AND HIGH-RESOLUTION AERIAL DATA COLLECTION IN REMOTE AND HAZARDOUS ENVIRONMENTS.

WHY IS THIS UAV NEEDED?

UNDERSTANDING THE REASON

Why a Hybrid VTOL UAV?

A purpose-built Hybrid VTOL UAV overcomes these limitations by providing autonomous aerial surveying with minimal human intervention.

✓ **No Runway Required**
- Vertical takeoff and landing from virtually any flat surface.

✓ **Zero Personnel Exposure**
- Remote autonomous operation keeps personnel away from hazardous environments.

✓ **Rapid Deployment**
- Fast mission setup and shorter survey cycles, reducing operational time and cost.

✓ **High-Precision Data Collection**
- Captures accurate geospatial, geological, and environmental data for better decision-making.

✓ **Long-Endurance Operations**
- Combines VTOL flexibility with fixed-wing efficiency for extended-range mining missions.



GEOMINER 1 SOLUTION

VTOL flexibility + fixed-wing efficiency + modular sensing + autonomous navigation

This enables rapid deployment and extended aerial surveying while reducing unnecessary personnel exposure to hazardous terrain.

Operational Challenges

1. INACCESSIBLE TERRAIN
- Remote mining areas and rugged terrain make conventional ground surveys difficult and time-consuming.

2. HAZARDOUS ENVIRONMENTS
- Mining sites can expose personnel to unstable ground, dust, toxic gases, water, and other operational hazards.

3. SURVEYING COST & TIME
- Conventional surveying requires significant personnel, equipment, and operational time.

4. UAV CAPABILITY GAP
- Multirotors provide VTOL capability but have limited endurance, while fixed-wing aircraft require suitable launch and recovery areas.

PROJECT OBJECTIVE

DESIGN A FEASIBLE, MODULAR HYBRID VTOL UAV

Develop and evaluate a 30 kg hybrid VTOL UAV capable of vertical takeoff and landing, efficient fixed-wing cruise, high-precision navigation, modular sensing, autonomous mission execution, and safe operation in remote mining environments.



CORE OBJECTIVES

1. Establish complete system requirements.
2. Develop the conceptual airframe.
3. Design the VTOL and cruise propulsion systems.
4. Design the avionics and electrical architecture.
5. Integrate the Pixhawk autopilot.
6. Configure ArduPilot QuadPlane flight control.
7. Use QGroundControl as the Ground Control Station (GCS).
8. Develop a modular payload system.
9. Perform aerodynamic and conceptual design analysis.
10. Evaluate the design through engineering simulation.
11. Analyze safety, redundancy, and reliability.
12. Select components using engineering justification.

ENGINEERING DESIGN WORKFLOW FROM REQUIREMENTS TO UAV CONFIGURATION

MISSION REQUIREMENTS
↓
SYSTEM REQUIREMENTS
↓
CONCEPTUAL CONFIGURATION
↓
Onshape CAD DEVELOPMENT
↓
OpenVSP AIRCRAFT MODELING
↓
XFLR5 AERODYNAMIC ANALYSIS
↓
PROPULSION & POWER SIZING
↓
AVIONICS & PIXHAWK INTEGRATION
↓
ARDUPILOT QUADPLANE CONFIGURATION
↓
QGROUNDCONTROL GCS CONFIGURATION
↓
SYSTEM INTEGRATION
↓
PERFORMANCE & SAFETY EVALUATION

SYSTEM REQUIREMENTS



COMPLETE SYSTEM REQUIREMENT SET

The proposed GEOMINER establishes these performance targets, including the 30 kg MTOW, 120-minute endurance, 100 km mission range, 5 kg payload and 2.8 m wingspan.

Category	*Requirement / Target*
Configuration	**Hybrid VTOL fixed-wing**
MTOW	30 kg
Payload	Up to 5 kg
Endurance	120 min
Mission range	100 km
Wingspan	2.8 m

Category	*Requirement / Target*
Cruise speed	**18–25 m/s**
Maximum speed	30 m/s
Hover T/W	≥1.8
RTK accuracy	<5 cm horizontal
Service ceiling	4,000 m ASL
Flight controller	Pixhawk

Category	*Requirement / Target*
Autopilot	**ArduPilot QuadPlane**
GCS	QGroundControl
Payload interface	Mechanical + power + data
Recovery	VTOL landing + emergency recovery
Mission operation	Autonomous with operator supervision

DESIGN TOOLS & SOFTWARE

ENGINEERING DESIGN & SIMULATION WORKFLOW

Onshape
Purpose:
- 3D CAD modeling
- Airframe design
- Component arrangement
- Structural layout
- Assembly development
- Design collaboration

SolidWorks
Purpose:
- Detailed 3D CAD modeling
- Part and assembly design
- Airframe and component development
- Structural design and validation
- Engineering drawings
- CFD and flow simulations
- Aerodynamic and fluid-flow analysis
- Design optimization and performance evaluation

XFLR5
Purpose:
- Wing and airfoil analysis
- Lift and drag estimation
- Aerodynamic performance evaluation
- Stability and performance assessment

ArduPilot / QuadPlane
Purpose:
- Hybrid VTOL flight-control architecture
- VTOL and fixed-wing flight modes
- Autonomous mission logic
- Navigation and failsafe configuration

QGroundControl
Purpose:
- Mission planning
- Parameter configuration
- Telemetry monitoring
- Ground-control interface
- Operator supervision

Pixhawk
Purpose:
- Primary flight-control hardware
- Sensor integration
- Navigation
- Actuator control
- Flight-state management

CONCEPTUAL DESIGN

INTELLIGENT MODULAR AERIAL MINING SYSTEM

KEY DESIGN FEATURES

1. HYBRID VTOL FLIGHT
Four lift motors provide vertical takeoff, landing, and hovering.

2. FIXED-WING CRUISE
A dedicated rear pusher motor provides efficient forward flight.

3. PRECISION NAVIGATION
RTK-GNSS, IMU, magnetometer, barometer, and airspeed sensing support navigation.

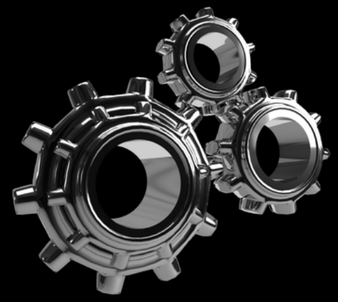
4. MODULAR PAYLOAD BAY
Quick-release interfaces allow different sensors to be installed for different missions.

5. ONBOARD COMPUTING
A companion computer provides data processing and future AI capability.

6. PIXHAWK FLIGHT CONTROL
Pixhawk serves as the primary flight-control platform running ArduPilot QuadPlane.

7. QGROUNDCONTROL GCS
QGroundControl provides mission planning, monitoring, configuration, and operator supervision.

OVERALL SYSTEM ARCHITECTURE



SEVEN MAJOR SYSTEMS & SUBSYSTEMS

1. STRUCTURAL SYSTEM

- Fuselage
- Wing
- Twin-boom tail
- Landing gear
- VTOL arms
- Motor mounts

Functions: Support, stability, aerodynamic performance, protection.

2. AVIONICS SYSTEM

- Pixhawk flight controller
- IMU
- RTK-GNSS
- Magnetometer
- Barometer
- Airspeed sensor
- Obstacle sensors

Functions: Stabilization, navigation, autonomy.

3. PROPULSION SYSTEM

- 4 VTOL motors
- 1 pusher motor
- ESCs
- Propellers

Functions: Lift, cruise thrust, transition.

4. ELECTRICAL POWER SYSTEM

- Main battery
- Backup battery
- BMS
- PDB
- Wiring and protection

Functions: Power supply, management, protection.

5. MISSION COMPUTING & COMMUNICATION

- Companion computer
- Telemetry
- Data storage
- Communication system

Functions: Mission computing, communications, data processing.

6. PAYLOAD SYSTEM

- Payload bay
- Quick-release mount
- 3-axis gimbal
- Mission sensors

Functions: Sensor integration, stabilization, data collection.

7. SAFETY & EMERGENCY SYSTEM

- Ballistic parachute
- Flight-termination system
- Locator beacon
- Geofencing

Functions: Recovery, emergency response, tracking.

FLIGHT CONTROL & GROUND CONTROL

PIXHAWK + ARDUPILOT + QGROUNDCONTROL

PIXHAWK AUTOPILOT

Primary flight-control functions

- Attitude stabilization
- Navigation
- Sensor integration
- Motor/servo control
- VTOL transition management
- Failsafe execution

ARDUPILOT QUADPLANE

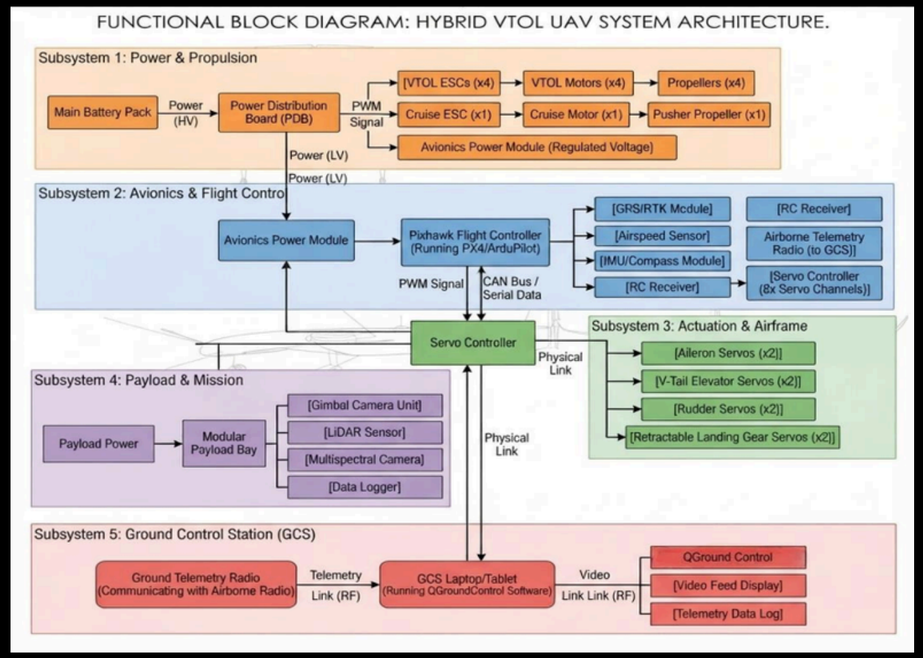
Used as the autopilot software for the hybrid VTOL configuration.

QGROUNDCONTROL GCS

Used as the ground control station for:

- **Mission planning**
- **Waypoint configuration**
- **Flight monitoring**
- **Parameter configuration**
- **Telemetry**
- **Operator commands**
- **Return-to-home supervision**

BLOCK DIAGRAM



Architecture:

QGroundControl GCs -- Telemetry Link -- Pixhawk -- ArduPilot QuadPlane -- Navigation / Propulsion / Payload / Safety

DETAILED AIRFRAME DESIGN

STRUCTURAL & AERODYNAMIC CONFIGURATION

PRIMARY STRUCTURE

- Carbon-fiber fuselage
- High-aspect-ratio fixed wing
- Twin-boom tail
- Reinforced landing system
- VTOL motor supports

DESIGN OBJECTIVES

LOW MASS

Reduce structural weight to preserve payload and battery capacity.

HIGH STIFFNESS

Maintain aerodynamic shape and flight stability.

MODULARITY

Provide accessible areas for avionics, batteries, payloads, and maintenance.

AERODYNAMIC EFFICIENCY

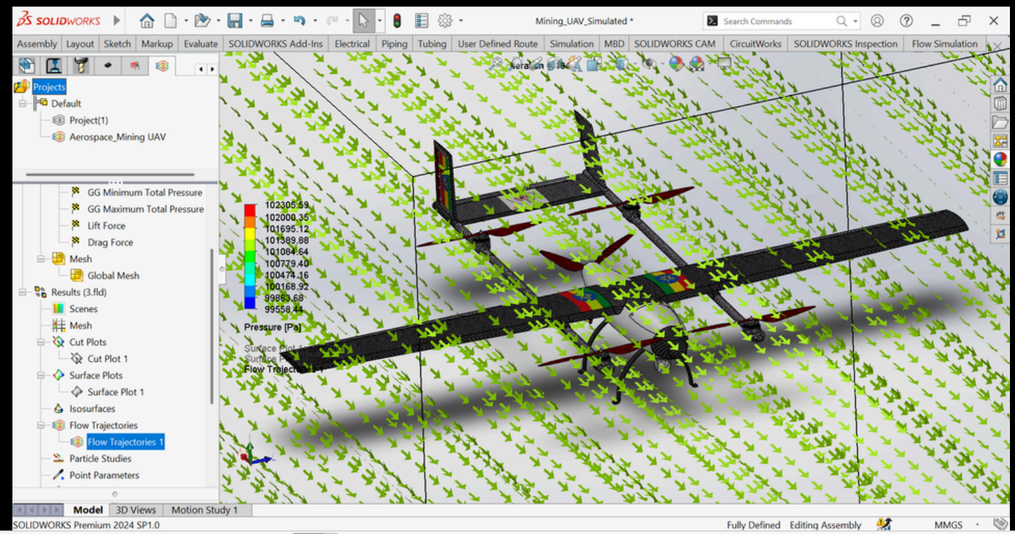
Optimize the aircraft for efficient fixed-wing cruise.

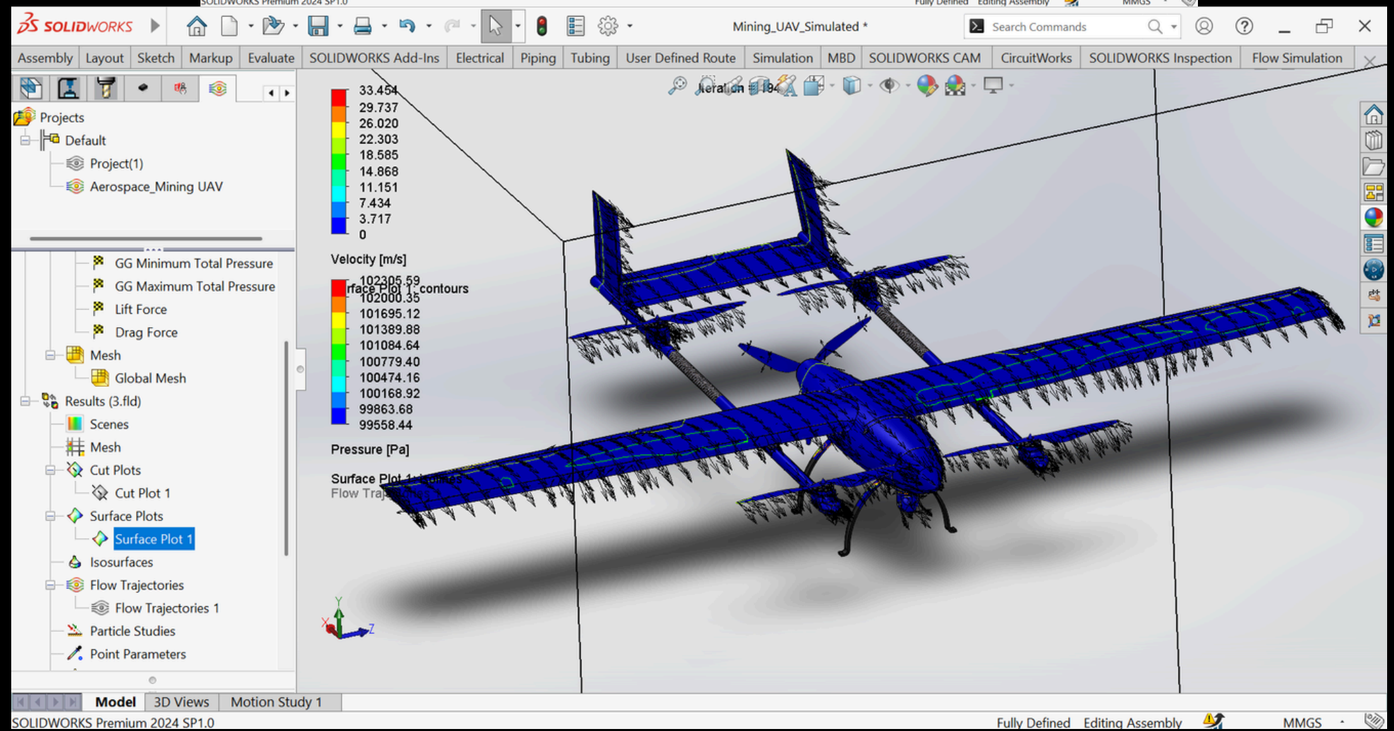
DESIGN SOFTWARE

Onshape → CAD / Assembly

SolidWorks → Detailed CAD / CFD & Flow Simulation

XFLR5 → Aerodynamic Analysis





COMPONENT SELECTION & ENGINEERING JUSTIFICATION

MAJOR COMPONENTS

Component	*Selection*	*Engineering Justification*
Flight Controller	**Pixhawk**	**Established UAV flight-control platform**
Autopilot	ArduPilot QuadPlane	Supports hybrid VTOL/fixed-wing operation
Navigation	RTK-GNSS	High-precision positioning
Structure	Carbon Fiber	High stiffness-to-weight ratio
VTOL Motors	4 × lift motors	Redundant vertical propulsion

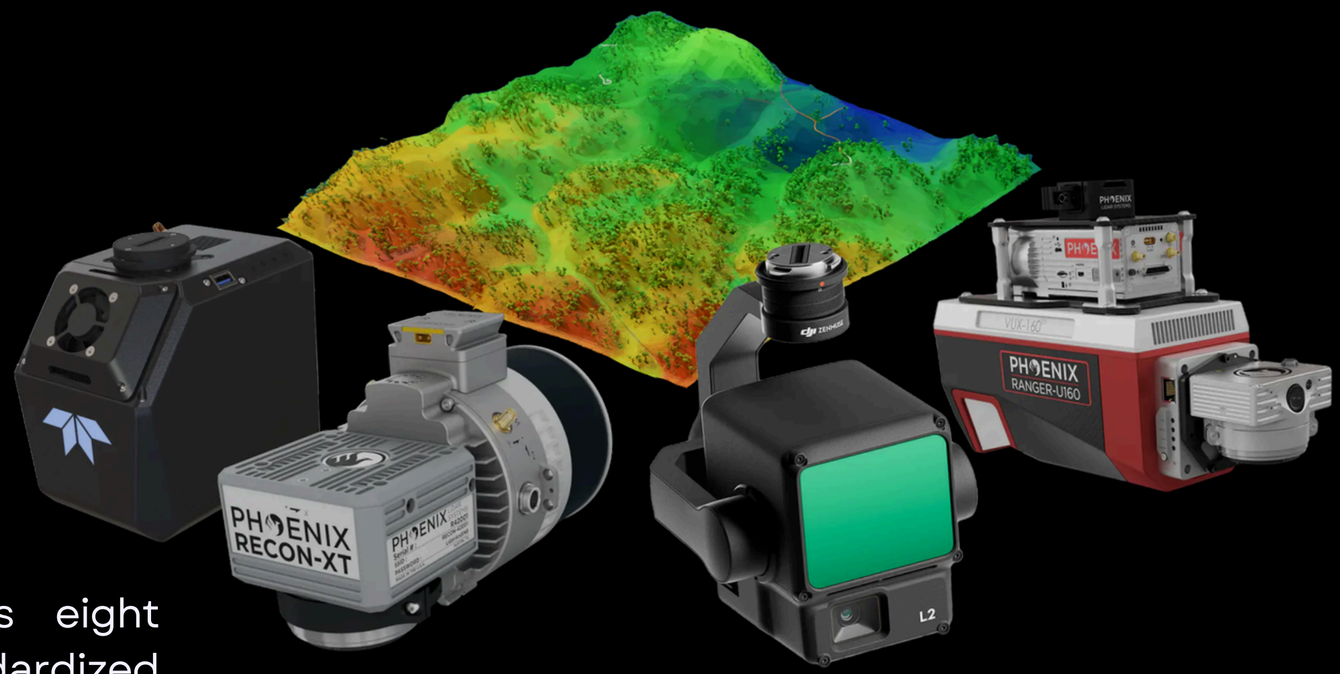
Component	*Selection*	*Engineering Justification*
Cruise Motor	**Rear pusher**	**Efficient forward cruise**
Battery	14S Li-ion	High energy density
Payload	Modular bay	Supports multiple mission sensors
Safety	Parachute + FTS	Emergency recovery capability
GCS	**QGroundControl**	Mission planning and flight supervision

MODULAR PAYLOAD SYSTEM

ONE UAV → MULTIPLE MINING MISSIONS

AVAILABLE PAYLOADS

- **Survey-grade LiDAR →** 3D terrain mapping
- **RGB mapping camera →** Orthomosaics
- **Hyperspectral camera →** Mineral identification
- **Multispectral camera →** Vegetation/land monitoring
- **Magnetometer →** Magnetic anomaly detection
- **Thermal camera →** Thermal inspection
- **Multi-gas sensor →** Hazardous gas detection
- **Radiation detector →** Radioactive-zone surveying



Maximum modular payload: 5 kg

The original GEOMINER proposal explicitly defines eight interchangeable sensor modules and a standardized mechanical, power, and data interface.

SIMULATION & ANALYSIS

ENGINEERING SIMULATION ACTIVITIES

SolidWorks
Completed / Used for:

- Detailed aircraft 3D modeling
- Wing and fuselage design
- Component and assembly development
- Structural design
- CFD and flow simulations
- Aerodynamic performance analysis
- Design validation and optimization

XFLR5
Completed / Used for:

- Airfoil analysis
- Wing aerodynamic analysis
- Lift/drag evaluation
- Preliminary aerodynamic performance assessment

QGroundControl
Intended for:

- Mission planning
- GCS configuration
- Flight visualization
- Autonomous mission setup
- Simulation/testing

Simulation Status:
Required aerodynamic and fluid-flow simulations have been successfully completed in SolidWorks. The UAV design is now prepared for software-in-the-loop (SITL) flight simulation using Gazebo, with the aircraft model to be integrated through URDF and connected to QGroundControl for flight-control and mission testing.

SAFETY, REDUNDANCY & MISSION RELIABILITY

LAYERED SAFETY ARCHITECTURE

FLIGHT SAFETY
- Geofencing
- Return-to-Home
- Emergency termination

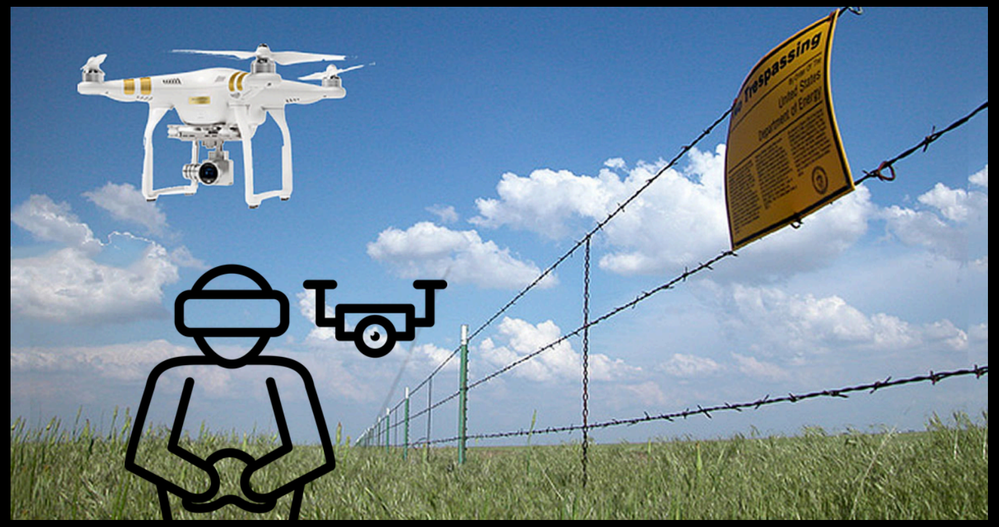
SENSOR REDUNDANCY
- IMU
- GNSS
- Barometer
- Magnetometer
- Airspeed
- Range sensing

PROPULSION REDUNDANCY
- Four independent VTOL motors
- Separate cruise motor

POWER REDUNDANCY
- Main battery
- Backup power

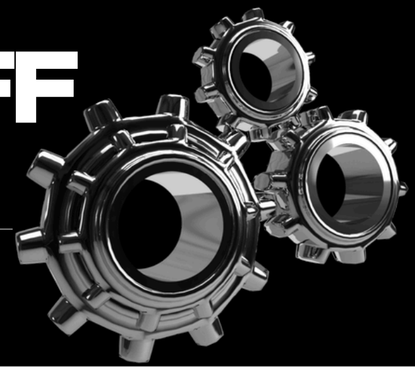
RECOVERY
- Ballistic parachute
- Locator beacon
- Emergency water-landing capability

DATA RELIABILITY
- Onboard data recording
- Data backup
- Telemetry monitoring





ENGINEERING TRADE-OFF ANALYSIS



KEY DESIGN DECISIONS

Trade-Off	*Selected Design*	*Reason*
Hover vs Endurance	**Hybrid VTOL**	**VTOL + efficient cruise**
Weight vs Strength	Carbon Fiber	High strength/stiffness-to-weight
Payload vs Endurance	Modular Payload	Carry mission-specific sensors
Battery vs Weight	Li-ion	High energy density
Speed vs Mapping	**Variable Cruise Speed**	Match mission requirements
Cost vs Performance	**Commercial Components**	Balance cost and reliability
Autonomy vs Safety	**Supervised Autonomy**	Human oversight for critical phases

EXPECTED DELIVERABLES

PROJECT OUTPUTS



1. COMPLETE PROJECT DOCUMENTATION
✅ Full engineering report
✅ PDF documentation
✅ PowerPoint presentation
✅ Simulation and analysis results

2. SYSTEM REQUIREMENTS
✅ Mission requirements
✅ Performance requirements
✅ Component requirements
✅ Airframe requirements
✅ Avionics requirements
✅ Payload requirements
✅ Safety requirements
✅ Simulation requirements

3. DESIGN & ENGINEERING
✅ Complete hybrid VTOL configuration
✅ Airframe design
✅ Propulsion system
✅ Electrical power system
✅ Avionics architecture
✅ Payload architecture
✅ Safety architecture

4. FLIGHT CONTROL
✅ Pixhawk autopilot
✅ ArduPilot QuadPlane
✅ QGroundControl GCS architecture
✅ Mission planning concept
✅ Autonomous flight-control architecture

5. ENGINEERING SOFTWARE
✅ Onshape – CAD design
✅ OpenVSP – aircraft configuration
✅ XFLR5 – aerodynamic analysis
✅ ArduPilot – autopilot architecture
✅ QGroundControl – GCS and mission-planning environment

6. SYSTEM ARCHITECTURE
✅ Complete system architecture
✅ Major systems and subsystems
✅ Functional block diagram
✅ Component interfaces
✅ Flight-control architecture

7. COMPONENT SELECTION
✅ Major component identification
✅ Engineering justification
✅ Weight and power considerations
✅ Performance considerations

8. SAFETY & RELIABILITY
✅ Safety analysis
✅ Redundancy strategy
✅ Failsafe architecture
✅ Emergency recovery
✅ Mission reliability considerations

9. PERFORMANCE
✅ MTOW analysis
✅ Payload capacity
✅ Endurance estimate
✅ Range estimate
✅ Propulsion sizing
✅ Aerodynamic analysis
✅ Expected operational capabilities

FUTURE IMPROVEMENTS & DEVELOPMENT ROADMAP

Future Improvements

- **AI & ML –** Onboard mineral detection and autonomous decisions
- **Digital Twin –** Real-time mine site modeling
- **Swarm UAVs –** Multi-drone coordinated missions
- **Satellite Communication –** Beyond-line-of-sight (BLOS) operation
- **Renewable Energy –** Solar-powered charging stations
- **Cybersecurity –** Encrypted links and GNSS protection





Development Roadmap

PHASE 1 – ENERGY

- Higher-density batteries and improved power management.

PHASE 2 – SENSING

- Advanced LiDAR, hyperspectral, thermal, and magnetic sensors.

PHASE 3 – AI

- Onboard mineral anomaly detection and intelligent data processing.

PHASE 4 – COMMUNICATION

- Satellite communications and extended-range operations.

PHASE 5 – MULTI-UAV

- Coordinated multi-UAV mapping of large mining concessions.



FINAL DESIGN SUMMARY

GEOMINER 1 is a professional Hybrid VTOL Mining UAV that integrates vertical takeoff and landing, efficient fixed-wing cruise, precision navigation, modular payloads, autonomous flight control, and layered safety systems for mineral exploration and mining operations.

Designed using Onshape, SolidWorks, and XFLR5, with SolidWorks used for CFD and fluid-flow simulations. Pixhawk + ArduPilot QuadPlane handles flight control, while QGroundControl provides mission planning and monitoring.

Project Status: The required aerodynamic and fluid-flow simulations have been completed in SolidWorks. The UAV is now ready for Gazebo-based SITL flight simulation, using a URDF model for integration with QGroundControl. The next stage is to validate flight dynamics, VTOL operation, transition, and autonomous control.



“ THANK YOU SO MUCH FOR YOUR ATENTION!!! ”

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